

Lab 1: Requirements Engineering & UML Use-Case Modelling

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PROBLEM STATEMENT #33: Micro-Lending & Peer Credit Risk Assessor

Requirement Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier (Low/Medium/High) is correctly assigned. Fail: An incomplete financial profile receives credit approval.	Helps assess the borrower's credit risk before lending.
FR-002	Functional	The system shall allow a peer	High	Pass: Lender can view the borrower's risk	Helps lenders make

		lender to review the borrower's credit risk rating and relevant risk information before lending.		rating before approving the loan. Fail: Risk information cannot be viewed.	informed lending decisions.
FR-003	Functional	The system shall generate a structured repayment schedule for an approved loan, including EMI amount and due dates.	High	Pass: The system generates an EMI amount and due date for every repayment. Fail: Any required EMI or due date is missing.	Provides a clear repayment plan for the borrower.
FR-004	Functional	The system shall process loan repayment transactions and update the borrower's	High	Pass: A successful repayment is recorded and the outstanding balance is reduced correctly. Fail: The payment is recorded without updating	Supports proper tracking of loan repayments.

		outstanding loan balance.		the balance.	
FR-005	Functional	The system shall trigger an automated default warning when a scheduled repayment becomes overdue.	Medium	Pass: A warning is generated when a repayment passes its due date without payment. Fail: No warning is generated for an overdue repayment.	Helps identify possible loan defaults early.
NFR-001	Nonfunctional	The system shall execute loan repayment transactions and EMI distribution inside database ACID transactions to prevent double-spending.	High	Pass: Testing confirms that concurrent transactions do not cause duplicate or inconsistent fund distribution. Fail: Duplicate or inconsistent transactions occur.	Ensures financial transaction integrity.
NFR-002	Nonfunctional	The system shall	High	Pass: Unauthorized users	Protects sensitive

		protect borrower financial informatio n from unauthori zed access.		are denied access to protected borrower financial information . Fail: Unauthoriz ed users can access the information .	financial and borrow er data.
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Functional Requirements

FR-001

Description: The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.

Priority: High

Acceptance Criteria: Pass: Risk tier (Low/Medium/High) is correctly assigned.

Fail: An incomplete financial profile receives credit approval.

Rationale: Helps assess the borrower's credit risk before lending. Type: Functional

FR-002

Type: Functional

Description: The system shall allow a peer lender to review the borrower's credit risk rating and relevant risk information before lending.

Priority: High

Acceptance Criteria: Pass: The peer lender can view the borrower's risk rating before approving the loan. Fail: The risk information cannot be viewed.

Rationale: Helps lenders make informed lending decisions.

FR-003

Type: Functional

Description: The system shall generate a structured repayment schedule for an approved loan, including EMI amount and due dates.

Priority: High

Acceptance Criteria: Pass: The system generates an EMI amount and due date for every repayment. Fail: Any required EMI amount or due date is missing.

Rationale: Provides a clear repayment plan for the borrower.

FR-004

Type: Functional

Description: The system shall process loan repayment transactions and update the borrower's outstanding loan balance.

Priority: High

Acceptance Criteria: Pass: A successful repayment is recorded and the outstanding balance is reduced correctly. Fail: The payment is recorded without updating the balance.

Rationale: Supports proper tracking of loan repayments.

FR-005

Type: Functional

Description: The system shall trigger an automated default warning when a scheduled repayment becomes overdue.

Priority: Medium

Acceptance Criteria: for passes its due date without payment. Fail: No warning

is generated an overdue repayment. Pass: A warning is generated when a repayment

Rationale: Helps identify possible loan defaults early.

Non-Functional Requirements

NFR-001

Type: Nonfunctional — Performance & Security

Description: Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.

Priority: High

Acceptance Criteria: Pass: Testing confirms that concurrent transactions do not cause duplicate or inconsistent fund distribution. Fail: Duplicate or inconsistent transactions occur.

Rationale: Ensures financial transaction integrity and prevents double-spending.

NFR-002

Type: Nonfunctional — Security

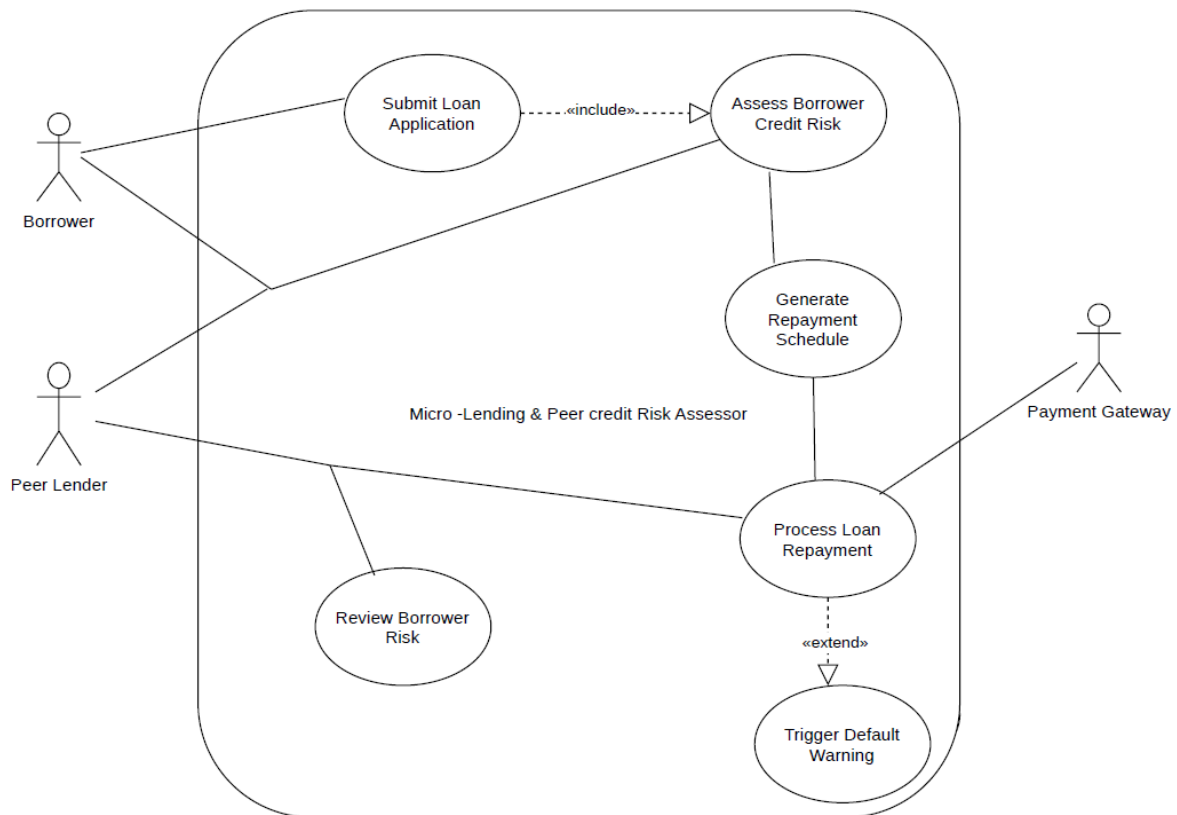
Description: The system shall protect borrower financial information from unauthorized access.

Priority: High

Acceptance Criteria: Pass: Unauthorized users are denied access to protected borrower financial information. Fail: An unauthorized user can access the information.

Rationale: Protects sensitive borrower and financial information.

UML Use-Case Diagram



Use-Case Flow Specification

Use Case: UC-04 – Process Loan Repayment

Primary Actor: Borrower

Supporting Actor: Payment Gateway

Preconditions

1. The borrower has an active loan.
2. A repayment/EMI is due.
3. The borrower has a valid payment method.

Main Success Scenario

1. Borrower selects the **Loan Repayment** option.
2. System displays the outstanding EMI amount and due date.
3. Borrower confirms the repayment.
4. System sends the repayment transaction to the Payment Gateway.
5. Payment Gateway authorizes the payment.
6. System processes the repayment using an ACID transaction.
7. System records the successful repayment.
8. System updates the borrower's outstanding loan balance.
9. System updates the repayment schedule.
10. System displays a repayment confirmation to the borrower.
11. The use case ends successfully.

Alternate Flow – Payment Declined

Payment is declined

1. Payment Gateway rejects the transaction.
2. System displays a payment-failed message.
3. System asks the borrower to try another payment method.
4. Borrower retries the payment.
5. If the repayment remains unpaid and becomes overdue, the system triggers a **Default Warning**.
6. The failed transaction does not reduce the outstanding loan balance.

Postconditions

1. Successful repayment is recorded in the system.

2. Outstanding loan balance is updated.
3. Repayment schedule is updated.
4. If the repayment becomes overdue, a default warning is triggered.